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RESEARCH ARTICLE

Can Artificial Agents be Authors?

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Abstract
Current Generative Artificial Intelligence models have become incredibly close to the human level of linguistic and artistic excellence, defying our conception of artworks as uniquely human products, resulting in an authorship problem, i.e., whether artificial agents can be regarded as genuine authors of their products. This paper provides a definition of institutional authorship to evaluate this possibility, using John Searle's Speech Act Theory and Theory of Institutions. To apply the definition, we focus on artistic cases, assuming the institutional theory of art. According to the institutional authorship definition proposed, authorship is the product of an agent's declarative acts in the context of relevant institutions granting the agent's deontic powers for performing such acts. These acts depend on sincerity conditions encompassing intentional states, which bind the product with an agent. We conclude that artificial agents cannot satisfy the institutional definition of authorship due to their inability to be institutional agents, meaning they cannot be authors.

Keywords Generative AI · Authorship · Speech acts · Institutional facts

1 Introduction

For a long time, artists have challenged the presupposition that there must be a causal or even intentional connection between an artist and an artwork.[1] However, the possibility of artworks being produced with some level of autonomy from human creators has become increasingly more real with recent Generative Artificial Intelligence systems (GAI)[2], as various models became available for public use. These are

[1] Examples include Marcel Duchamp's ready-mades, such as 1913 *Bicycle Wheel* or the 1917 *The Fountain*, and John Cage's 1951 random composition *Imaginary Landscape No.4 (March No. 2)*, an attempt to separate the artists' intention from the production process. See Cage (2013, p.36;57–60).

[2] Prior examples of mechanical or computer generated artworks are Jean Tinguely's 1959 *Méta-matic no. 10* and David Cope's *Experiments in Musical Intelligence*. See Boden (2007) for a defense of Cope's work.

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systems taking text prompts as inputs to generate different types of outputs, either texts, images, sounds, or video. Examples are the text-to-text ChatGPT based on OpenAI's GPT large language model, the text-to-image Dall-E and Midjourney, the text-to-sound music generator Suno, and the text-to-video SORA model.

These GAI systems are incredibly close to the human levels of linguistic and artistic excellence and defy our conception of artworks as uniquely human products. Numerous concerns have been raised on the practical and ethical consequences of such systems, spanning from educational misuses of AI-generated texts by students, to copyright infringements coming from unlicensed use of data for training. But above all, their widespread use challenges us with an *authorship problem*, i.e. whether artificial agents, such as the GAI systems, can be regarded as genuine authors of their products. In 2022, the AI-generated artwork *Théâtre d'Opéra Spatial*, created by Midjourney, won the Colorado State Fair's annual art competition, generating, as expected, a negative repercussion from other artists (Roose, 2022).[3] A consequence of the authorship problem on AI concerns copyright matters, which is especially the case for generative models. These models use a large database to generate their images/texts, which are also dependent on users' prompt commands. However, copyrights of the final product are still a matter of controversy.[4] If humans cannot be the authors of GAI systems' outputs, can the systems be? The possibility of artificial agents being authors of autonomous speech or creative production of artworks depends on their capacity for genuine agency, and regardless of the excellency that GAI systems are showing, it is still debatable whether they currently satisfy the relevant agency for authorship.

My aim in this paper is to provide an answer to the question of whether artificial agents can be authors. The argument runs as follows. Restricting the discussion to artistic cases, I first discuss some recent theories of artistic authorship (Section 2) to then propose an *Institutional Definition of Authorship* (Section 3). The definition is based on the theoretical landscape of John Searle's theory of Speech Acts and Social Institutions. To apply this definition to artistic cases, I assume the Institutional Theory of Art advanced by Catharine Abell. Generally, institutional authorship is considered the product of an agent's declarative acts in the context of relevant institutions that grant the agent's deontic powers for performing such acts. These acts depend on sincerity conditions encompassing intentional states, which bind the product with an agent (the artwork with its corresponding author). In Section 4, this definition is put to use concerning artificial agents, particularly GAI systems. My conclusion is that current GAI systems are not capable of satisfying the definition and, therefore, they currently cannot be authors. At the same time, the institutional theory leaves open the route for using the GAI system's outputs in artists own works, as similar cases were already accepted within the artworld.

[3] In another case, the Brazilian literary award *Prêmio Jabuti* excluded a contender for the 2023 best book illustration award after discovering the artist's use of Midjourney (Queiroz and Guerra, 2023).

[4] The U.S. Copyright Office rejected the artist's claim of authorship for *Théâtre d'Opéra Spatial*, as according to the copyright guidelines issued by the Office (2023), only products of human creativity are copyrightable. See also Wilson (2023) for the judge's decision on the matter.

2 Theories of Artistic Authorship

It is common sense that any account of authorship must include a causal and intentional component. As Uidhir (2013) explains, most theories assume authorship to be a two-placed relation *xAy*, ''*x* is the Author of *y*'', where ''[…] the author-relation is a causal-intentional relation between an agent (actual or non-actual) and the substantive products of that agent's intention-directed activities, such that, *xAy* only if *y* is the substantive product of an intention-directed activity for which *x* is the source.'' (p.47–48)[5]

Given that not every product of an intention-directed[6] activity is prone to be authored, Uidhir (2013) argues in favor of a three-place relation *xA*(*Fw*), read as ''*x* is the author of *w* as an *F*'' (*F* being a sortal description). As he claims, for an author *A*, a work *w* and sortal *F*, ''*A* is an author of *w* as an *F* if and only if *A* is directly responsible, at least in part, for *w* as an *F* (i.e., the way in which *w* falls under sortal *F*).'' (p.49) The main reason to adopt a three-place relation is that not every intentional activity defines authorship. Those *F*'s that can be authored are called *author-relevant F*'s.[7] There are two cases of author-relevance considered, according to the extent to which satisfying the *F*-work description is dependent on the agent's intention-directed activities, or the *F*-attempt (Uidhir, 2013, p.51):

> *Weak Author-Relevance*: *F* is *weakly* author-relevant just in case for all works *w*, *w* being an *F* merely *prima facie* entails *w* being an *F*-work [*w* as an *F* *prima facie* has an author].
>
> *Strong Author-Relevance*: *F* is *strongly* author-relevant just in case for all works *w*, *w* being an *F* (necessarily) entails *w* being an *F*-work [*w* as an *F* must have an author] (Uidhir, 2013, p. 51).

[5] Hilpinen's (1993) account of artifacts is one example.

[6] The term ''intention'' is ambiguous, and following Anscombe's influential book (1963), it can be understood either as an intentional action (doing something with some goal, that is, an event), the intention to do something in the future (a mental state), or the intention with which an action is performed (a reason). Of course, solely having a reason for producing something does not suffice for an artwork to exist (it could be argued that even extreme cases such as 1991 ''Show in Advance of Its Existence'' by Henry Flynt are intentional actions, not just future intentions). To be intention-dependent, an artwork must be ''the product of an intentional action'' (Uidhir, 2013, p.23). At the same time, not every product of an intentional action is an artwork. Thus, ''Artworks are the products of the attempts in which we engage, and intentions figure substantively by directing the actions or activities constitutive of those attempts'' (*ibidem*). Thus, by intention-dependent is meant the product of an intentional action (as an event) coupled with a substantive role played by the intention (as a mental state) orienting the artists'action. Explaining this connection between the intention to act and the act itself is out of my bounds. But a clear advantage of assuming the Searlean account is that ''Intentional states represent objects and states of affairs in exactly the same sense that speech acts represent objects and state of affairs'' (Searle, 1979b, p.75). Thus, we can have speech acts as proxys for intentional actions, and speak about intentional states as the sincerity conditions for those acts.

[7] Some *F* descriptions cannot be authored because they are not descriptions of any entity resulting from the work of an agent, or an *F*-work (e.g. being a *proton*). But some *F*-work descriptions cannot be authored because they are not *F*'s that are considered author-relevant. This is Uidhir's case against Hilpinen's (1993) account. By being too broad, it includes any man-made artifacts in the possible relation to an author, meaning that a hammer would have an author in the same sense a novel has an author. Clearly, hammer is not an author-relevant description, whereas a novel is.

For strong author-relevance cases, there cannot be a *w* that is an *F*-work without entailing the existence of an author, since *w* satisfies the *F*-work precisely because the intention-directed activities of an agent. But for weak author-relevance cases, *w* being an *F* only *hints* to the existence of an author, given that satisfying the weak *F* description does not necessarily requires such intention-directed activities. Suppose *w* satisfies the sortal *poem*. It could be the case that literary conventions and relevant institutions consider that if *w* is a poem, then *w* must have an author, i.e. it must be the production of some intention-directed activity of an agent. In such case, *poem* is strongly author-relevant. But conventions and institutions may change, making a *poem* to be only weak author-relevant, such as with computer-generated lines of texts that *looks like* to be a poem. In this case, if *w* is a poem, it only *prima facie* entails that *w* might be the product of a poem-attempt, viz., the product of an intentional activity that an agent is responsible for, meaning that *w* only *prima facie* has an author.[8]

If this is right, the possibility of computer-generated artworks, such as the ones produced by Generative AI models, having an author depends (among other conditions) on the sortal description *artwork* being either weak or strong author-relevant for these cases. Not only that, it will also depend on the possibility of such systems having intentional states showing that it is in the model's interest to produce a *w* that satisfies the sortal *artwork*. A plausible possibility is to consider *artworks* as weak author-relevant in the cases involving generative AI productions. If so, it would suffice to say that if a product of a generative AI model *w* only *prima facie* satisfies the sortal *artwork*, then it only *prima facie* is an artwork and only *prima facie* has an author. Rephrasing Uidhir (2013, p.53), if it looks like an artwork, then it suffices to say that it looks like it has an author.[9] However, it is doubtful that the sortal *artwork* is weak author-relevant in all cases, e.g. in Fine Arts. What the author-relevance property shows is that authorship is dependent on factors that go beyond intentional acts, and thus, the answer to whether GAI systems can be authors must go beyond their ability to represent intentional states.[10]

Even automated productions are somewhat intentional, as there is a level of intentional activities involved. Consider, for instance, OpenAI's large language model GPT and its chatbot ChatGPT. Any given prompt product of ChatGPT is intentionally dependent at least on OpenAI's engineers and the user behind the prompt

[8] If *poem* is weak author-relevant, then cases of authorless poems are allowed, even though they are still expected to have one. If *w* is *prima facie* a poem, then it *prima facie* have an author.

[9] But it would still be questionable whether the model produces a *w* with the intention of satisfying the sortal *artwork* in the first place. This could be met with Dennett's Intentional Stance (1989), where it would suffice to say that if it looks like it has intentional states, then it looks like to be intentionally responsible for *w* to be an artwork. We will return to this point in Section 4.1.

[10] Hick (2017) similarly argues that intention alone is not sufficient for explaining authorship. In his view, authorship ''[…] consists in the exercise of one's power to select and arrange elements as constitutive of that work'', where ''the rules of the art game determine what makes an individual an author of a given work'' (p.76). These rules are determined by historical, sociological and anthropological factors. To say that an author *A* has the recognized social and institutional powers to be the author of an *F*-work must entail that *F* is author-relevant in Uidhir's sense. Thus, Hick's condition entails author-relevance.

command,[11] as these are the closest intentional links to the final product. Based on this assumption, Vlaad (2024) argued in favour of a conjunctive account of authorship for AI generated images, where authorship is attributed both to the developers and the prompt-giver. A similar answer was already suggested by Hilpinen (1993, p.157): ''It is obvious that objects which are produced automatically, by machines […], should be regarded as intended (or 'intentional') products; such objects are authored by the persons who designed the machines or control their operation.'' However, this seems to miss the point of authorship.[12] It seems unreasonable to consider OpenAI as co-author of every GPT output, just as it is unreasonable to consider Leo Fender the co-author of every song composed with a Fender electric guitar.

3 Institutional Authorship

Following Uidhir's account of authorship as a three-place relation, my hypothesis is that most of the relevant cases concerning authorship, viz., those satisfying the strong author-relevance property, involve what I will call *Institutional Authorship*. According to the institutional account, an agent *A* is the author of a work *w* as an *F* just in case:

1. *w* results from an act performed by *A* having the logical form of a declaration,
2. the content of *A*'s act includes that w is an *F*-work,
3. *w* is accepted as an object of type *F* and
4. *A* has the required (deontic) powers to realize an *F*-work.

The theoretical landscape for this account comes from John Searle's theory of Institutions (1995; 2010) and Speech Act Theory (Searle, 1979a; Searle and Vanderveken, 1985). Hence, the definition is thought to capture authorship concerning those *F*'s that are institutionally defined. In order to evaluate authorship for generative AI models on an artistic level, we want to restrict the *F*-objects to be *artworks*. Hence, another theoretical presupposition is the Institutional Theory of Arts, specifically the one advanced by Abell (2012). Given that it is not my goal to defend such theory, it should be clear that my argument (precisely the application of the institutional definition to artistic cases) is premissed on the acceptance of the institutional theory of art. Even without it, the definition should still be applicable to any institutional *F*-work.

Since the successful conditions for declarative acts include some intentional states on the part of the speaker, it follows that condition (1) already encompasses

[11] One could also include the dataset on which GPT was trained as intentionally relevant, but although the dataset seems to be influential, it is not in an author-relevant sense.

[12] Even granting the point that collective authorship is the best explanation for Generative AI models' productions, we would still have a negative answer to the possibility of artificial agents being authors.

the intentionality required for authorship. Condition (2) states that the propositional content of the declarative act in (1) must be that *w* satisfy the *F*-type. Therefore, conditions (1) and (2) together state that the agent *intends w* to be an *F*, that the object is an attempt to match his intentions. Conditions (3) and (4) specify the broader context under which the object-type *F* is regimented and where *A*'s position is evaluated.[13] Notice also that whether *w* satisfy *F* or not is something to be evaluated within the *F*-community, as it specifies which properties are relevant for *F* and which agents are *F*-relevant.

A corollary of this definition is that all *F*'s that are institutionally relevant are also strong author-relevant in Uidhir's sense. This holds because institutional authorship requires an act having the logical form of a declarative speech act, which in turn requires an agent to be performed. Since declarations are author-dependent activities, all *F*s that are institutionally defined are also strong author-relevant.[14] Thus, any *F*-work satisfied by the institutional definition of authorship also satisfies the strong author-relevance property, meaning that *F* has an author. In other words, it is easier to evaluate authorship on strong author-relevant *F*'s than on weak ones, hence the definition's restriction to only institutional *F*'s.

Recently, van Woudenberg et al. (2024) presented a similar argument for authorship, particularly on OpenAI's chatbot ChatGPT. They propose a conservative argument, relying on a standard conception of agency with proper intentional states. Accordingly, they conclude that ChatGPT cannot be an author as it cannot perform illocutionary acts such as assertions and promises. As I shall explain, the institutional definition here proposed agrees with the conservative view and can be considered complementary due to its different scope. Focusing on declarative cases, the institutional account is broader and offers an explanation for all authorship cases involving institutionally dependent artifacts.

3.1 Speech Act Theory

Speech Act Theory is roughly the study of what we can do with words and propositions. For a given propositional content *P*, we may assert that *P* is the case, ask whether *P* is the case, declare *P* to be the case, etc. Language has special ways of

[13] The deontology relative to an agent is certainly defined over a collective framework. However, the institutional definition of authorship has the drawback of assuming only those kinds of speech acts that have an individual speaker, a natural consequence of the Searlean account assumed. Collective speech acts are fairly common in linguistic contexts (e.g., ''we, the jury, declare the defendant not guilty'') and artistic authorship cases. But according to Meijers (2007), they are *sui generis*, not reducible to speech acts with individual speakers. However, collective speech acts can be adjusted to Searle's theory of speech acts, as his theory of social institutions already largely depends on the phenomena of collective intentionality, meaning that collective declarations are already constitutive of the institutional reality. Nonetheless, the extension of the institutional definition of authorship will not be pursued in this paper.

[14] Consider the extreme case of found art. A found art is not an artwork that is found, but a found object that is *considered* (viz., declared) as an artwork, meaning that it requires an act that has the logical structure of a declarative. Hence, even found art must have authors.

realizing each of these different illocutionary acts, called *illocutionary force indicators* (like verbs, expressions, and all sorts of intonations).[15] For any proposition *P*, *F(P)* denotes the illocutionary act in which *P* is expressed with illocutionary force *F*. Searle and Vanderveken (1985) considers seven important dimensions for an utterance having this structure. The *illocutionary point*, which marks the intent of the act, is the most important component of illocutionary force.[16] Another important aspect of a speech act is the associated *sincerity conditions*, that is, the psychological state that is expressed by the linguistic act. By asserting that *P* a speaker is expressing the belief that *P* is true, while by directing a hearer to do *P* or by declaring it that *P*, one is expressing the desire for *P* to be performed or to be the case.

The performance of a speech act may also require some particular *modes of achievement* to be uttered. For example, an order can be uttered by invoking a position of authority. But for the utterance to be successful, there are also *preparatory conditions* to be met. These ''[...] are defined for each context of utterance *i* and proposition *P*, which states of affairs the speaker a_i must presuppose to obtain in the world of the utterance W_i if he performs the illocution *F*(*P*) in *i*.'' (Searle and Vanderveken, 1985, pp.43–44) Thus, the mode of achievement of an order can only be successful if the speaker indeed has the required authority.[17]

Searle (1979a) and Searle and Vanderveken's (1985) resulting taxonomy have five classes: *assertives*, *directives*, *commissives*, *expressives* and *declaratives*. Our point of interest is with declaratives. They have the illocutionary point (formalized with a capital *D*) of attempting to alter the world, but it differs from other speech acts in virtue of changing the world simply by representing it as being changed (if successful). It has a double direction of fit (world-word) and a double sincerity condition: belief and desire. For a propositional content *P*, declarative acts have the following structure: $D \updownarrow b/d\ (P)$. Finally, in order to successfully utter a declarative with such structure, the following conditions are necessary, as explained by Searle and Vanderveken (1985, p.57):

> All declarative illocutionary forces have the mode of achievement that the speaker invokes his power or authority to perform the declaration and the general preparatory condition that the speaker has that power or authority to change the world by the performance of the appropriate utterance act.

According to conditions (1) and (2) of the institutional authorship account above outlined, an authored work *w* is the product of a declarative act in which the content

[15] The first study on the taxonomy of illocutionary forces and speech acts comes from Austin (1962). Here we follow exclusively the one advanced by Searle (1979a) and Searle and Vanderveken (1985).

[16] For example, the point of an assertion is to tell how things are. The point of a declaration is to declare something to exist.

[17] The remaining components of illocutionary forces that we shall not assess here are the degree of strength of the illocutionary point, the propositional content conditions, and the degree of strength of the sincerity conditions. Additionally, there is an underlying relationship between the propositional content and what Searle and Vanderveken (1985, p.28) call the world of utterance. This is the *direction of fit* between the world and the words. There are four possible directions: (1) *word-to-world* (↓), (2) *world-to-word* (↑), (3) the *double* (↕) world-word direction, where the speaker attempts to alter the world by representing it as being so altered; and (4) a *null* direction.

P states that *w* is an *F*-work. Thus, the act-structure of these declaratives is roughly *D* ↕ *b/d (w is an F)*, where an agent expresses the belief that *w* is an *F* and the desire to make *w* an *F*. Furthermore, the agent must have the necessary power or authority to perform the intended change in the world, relative to the *F*-type, *per* preparatory conditions. If *F* is the sortal *artwork*, by declaring *w* to be an artwork, an agent is expressing the belief and desire that *w* is an artwork.

It is important to remember that by declaring it as such, an agent is *ipso facto* making it the case that *w* is an *F*. But in order to successfully declare a *w* to be an *F*, the agent must be in a position of having the contextually relevant powers that are specified by the *F*-community. These are specified in conditions (3) and (4) of the institutional definition of authorship outlined. These are institutional conditions, making the nature of the resulting product of the declarative act an institutional entity, as we shall see.

3.2 The Theory of Institutions

Above and beyond the realm of brute facts and objects lies a realm of institutional facts that we take for granted in our daily lives. This realm is populated by governments, universities, money, marriages, chess games, wars, and everything in between that exists by virtue of social agreements, and according to John Searle's theory of institutional facts (1995; 2010), are entities defined by systems of constitutive rules.

A given practice is put forward by a constitutive rule if the rule is *constitutive* of the practice. The rules of chess not only regulate the behavior of the players, but they constitute chess as a game. The basic structure of constitutive rules is ''*X* count as *Y*'' or ''*X* count as *Y* in context *C*'', and according to Searle, all institutional forms of human culture have this structure (Searle, 1995, p.40). Constitutive rules are also the decisive step from a physical to a social reality as they provide ''[…] collective intentional imposition of function on entities that cannot perform those functions without that imposition'' (Searle, 1995, p.41). Money, as printed currency, cannot perform its institutional function without such function being constitutively imposed. It is not a piece of paper that has value; money has.

The imposition of function also imposes a new *status* to the entity, and thus constitutive rules in the form ''*X* count as *Y* in *C*'' are also called *status functions* by Searle. As he explains,

> the application of the constitutive rule introduces the following features: The *Y* term has to assign a new *status* that the object does not already have just in virtue of satisfying the *X* term, and there has to be collective agreement, or at least acceptance, both in the imposition of that status on the stuff referred to by the *X* term and about the function that goes with that status. (Searle, 1995, p.44)

And in order for this new function to be carried over by the *X* entity, there must be ''continued collective acceptance or recognition of the validity of the assigned function'' (Searle, 1995, p.45).

One way to impose such functions is by making performative utterances, viz., declarations, as ''[…] the status-function marked by the *Y* term in the formula '*X* counts as *Y*' can often, though not always, be imposed simply by declaring it to be imposed'' (Searle, 1995, p.55). Searle (2010, p.13) also claims that ''all of human institutional reality is created and maintained in existence by (representations that have the same logical form of) SF Declarations, including the cases that are not speech acts in the explicit form of Declarations''. The most common cases are those in which an agent makes the case that *X* functions as *Y* by explicitly declaring it to be a *Y*.[18] But sometimes a new function may be declared implicitly by an agent by simply behaving *as if X* functions as *Y*.[19] In artistic cases, painters can make explicit declarations by signing the painting and giving it a name. They can also make implicit declarations by simply acting as if the artwork is finished, therefore making the case that it is an artwork having the artist as the author.[20]

Finally, the imposition of status functions also includes the imposition of *deontic powers*, i.e., ''rights, responsabilities, obligations, duties, privileges, entitlements, penalties, authorizations, permissions'', etc. associated with the new status. (Searle, 1995, p.100). The imposition of the status ''money'' to some pieces of paper endows the one carrying it to exchange for equal value, but also the duty to pay taxes relative to each monetary transaction. In some cases, the possibility to perform declarative acts and thus to impose new status functions are also included within such powers. For example, making someone a university professor is to entitle him/her to a series of rights, responsibilities, permissions, etc., including the power to approve/fail a student. Similarly, to ordain someone a priest is to entitle him to impose new status functions, e.g. to perform a marriage. These cases reflect the fact that ''status functions depend on collective intentionality'' (Searle, 2010, p.8), meaning that the whole structure of constitutive rules, status functions, and deontic powers is dependent on collective recognition.

With that in mind, we can better assess the conditions for institutional authorship above. Since condition (1) requires *w* to have the logical form of a declarative act and condition (2) requires the content of such act to state that *w* is an *F*, it follows that by authoring *w* an agent is in fact imposing the *F*-status to the entity *w* (that *w* count as an *F*). And because such acts are dependent on the structure of social institutions, the *F*-status is an institutional status, given that to be an *F* is contextually tied to some relevant institutions that specify what it is to be an *F*, therefore fulfilling the condition (3). Finally, the institutional status relative to *F* also includes

[18] The rules of chess, the declaration of independence, marriages, etc. are standard examples.

[19] I can make it the case that this piece of wood is a stool by simply sitting on it.

[20] We can read this imposition of a *status* precisely as the declaration's attempt to alter the world, which renders *all* institutional entities as successful attempts. If the institutional theory of art is accepted, then likewise all productions of works of art are attempts to impose *status functions* and thus to alter the world. This should apply to texts, novels, paintings, music compositions, etc. For instance, we can say both that ''this set of ordered written-tokens of text *counts* as Don Quixote'' or ''this set of ordered musical notes (either written or performed) *counts* as Beethoven's 5th symphony''. Therefore, the scope of objects being authored is precisely those that can be introduced by means of status functions, constitutive rules, and declarative acts. However, the claim that artworks in general fall under this description should still be flagged as an assumption, one that I shall not pursue in more detail.

other status and deontic powers that are relevant for realizing an *F*, thus specifying condition (4). For example, a judge may be considered the author of a verdict if (1) the verdict results from an act performed by the judge with the logical form of a declaration, (2) the content of the act includes that the work being produced is a verdict, (3) the work is accepted as a genuine verdict (that is, it satisfy the conditions for being a verdict under the relevant legal community), and finally, (4) the judge has the required deontic power to issue a verdict (otherwise, it would not be a successful declarative act, failing condition (1)). If all four conditions are met, the judge is successfully the author of the verdict.

3.3 The Institutional Theory of Art

According to the institutional account above sketched, conditions (1)–(4) are dependent on the *F*-type being evaluated. This is the case since the network of deontic powers is context-dependent on the *F*-type. Therefore, evaluating the possibility of Generative AI systems being authors is dependent on each *F* case. For now, we want to restrict the consideration to artistic cases, that is, to the sortal description *artwork*, and thus, one must consider artworks on an institutional basis.[21]

The first definition of artworks aiming for a context-dependent answer came with Arthur Danto's *Artworld* (1964). As he claimed, ''To see something as art requires something the eye cannot decry - an atmosphere of artistic theory, a knowledge of the history of art: an artworld.'' (p.580). What he meant by an artworld is, roughly, a realm beyond the ''real'' world, establishing ''a fundamental contrast between artworks and real objects'' (p.582). Without qualification, this is nothing more than the class of all artworks, and the same could be said about the class of all *F*-objects that they form a *F*-world. But Danto is also focusing on the *history of art* as informing us what is or is not inside the artworld. For Adajian (2022), it is the reliance on the historical context that makes Danto's theory an institutional theory, although he did not claim to be providing one.

An explicit institutional account was then famously proposed by George Dickie.[22] In Dickie's definition, artworks are defined within ''the network of relations or context within which a theory places works of art,'' which he calls a *framework* (Dickie, 2019, p.16). The gist of his theory is that ''A work of art is an artifact of a kind

[21] At this point, it is questionable whether the description *artwork* defines a precise class of things or whether we should call artworks a sortal description. Assuming the institutional theory of art, we can argue that this class ideally includes paintings, novels, musical compositions, and all things introduced through declarative acts and constitutive rules and that are collectively recognized as artworks. Thus, collective agreement is in principle capable of satisfying some of the criteria for sortals, that is, for identity, counting, and continued existence of its objects. But, one may still object, collective agreements are inherently a vague notion. To my mind, there is no way other than to bite the bullet, and accept that the class of artworks may not have sharp boundaries. Searle (1995) argues that institutional facts are both ontologically subjective and epistemically objective. If that's acceptable, then the institutional theory of art may find its value precisely by making the class of artworks an ever-changing reality (by being ontologically subjective) that is still immune to relativistic interpretations (by being epistemically objective).

[22] Dickie's theory has undergone multiple versions, and it is not my goal to assess them all. For brevity reasons, the version mentioned is the revised and summarized version found in Dickie (2019).

created to be presented to an artworld public'' (Dickie, 2019, p.20), where an artworld public is formed by those members of the artworld that share with the artist the knowledge and understanding relevant to the framework. What makes Dickie's theory institutional is the condition that artworks are ''the result of the position or place they occupy within an established practice'' (Dickie, 2019, p.18), meaning that what makes an artwork is dependent on the position that it fulfills within relevant social institutions.

Abell (2012) offered an approach to the institutional definition of art based on Searle's account of social institutions. According to Searle, the existence of an institution depends on the function that it is assigned to perform. Universities exist to promote higher education; the police exist to promote safety to the community in which it is based. However, since institutions exist by virtue of collective agreements, their existence depends solely on participants accepting that it fulfills such a function, not on their actual capacity to effectively perform it.[23] Similarly, according to Abell (2012, p.683) ''the existence of an art institution may be due to its being perceived to perform a variety of different functions'', according to different subgroups within the art institution and time.[24] Nonetheless, there is no need to take such functions as constitutive of the definition of artworks. Since these functions are assumed within the context of an institution, what is needed is just the recognition that artworks are institutional-dependent. In fact, functions may vary according to contingencies of the institutions, a condition that is consistent with the vagueness of the term ''art'', according to Abell.

According to this definition, that ''*X* count as an artwork in *C*'' is conditioned on the institutional structure of *C*, involving both facts and deontic powers relative to the agents that are recognized in *C* and facts about artworks themselves. This involves conditions (3) and (4) of institutional authorship above sketched. Some artworks may be easily accepted within the institutional framework by virtue of satisfying the conditions for being an artwork, and in such cases, the acceptance is less dependent on condition (4). On the other hand, works that are considered disruptive may benefit from their authors being already accepted within some institutional framework in order to successfully be accepted as an artwork. Certainly, Duchamp's attempt to exhibit the *Fountain* would have been much more difficult hadn't he already been a recognized artist in the European artworld. Hence, the institutional structure involves different kinds of functions, deontic powers, and institutional facts that together entail what an artwork is.

Abell's definition is then summarized as: ''Something is an artwork iff it is the product of an art institution, and it directly affects how effectively that institution performs the perceived functions to which its existence is due.'' (Abell, 2012,

[23] The university may fail to fulfill its goals in particular cases, or the police may be responsible for the abuse of power, but they are thought of as institutions for education and safety nonetheless.

[24] Examples of functions that an art institution may perform include ''promoting positive aesthetic properties; promoting the expression of emotion; facilitating the posing of intellectual challenges; promoting formal complexity and coherence; facilitating the communication of complex meanings; promoting the exhibition of individual points of view; promoting originality; and promoting the exercise of a high degree of skill.'' (Abell, 2012, pp.683–684).

p.686) One upshot of this definition is that it doesn't have anything to say about how an artwork comes into being. ''To say that something is the product of an art institution is not to say anything about how it came to have its physical composition or structure'' (*idem*.). We can explain Duchamp's *Fountain* or Picaso's *Guernica* as artworks by institutional recognition, even though each artwork is accepted by different standards.[25]

What is needed to put forward such artworks within the institutional framework is, per condition (1) of institutional authorship, an act that has the logical form of a declarative. This position is also argued by Davies (1991). Abell (2012, p.676–677) differs on grounds that the authority that confers someone the ability to perform declarative acts is contingently determined, meaning that a definition of artworks cannot rely on such acts. However, the contingent character of authority is nothing more than a feature of how institutions are historically determined and a crucial factor in explaining how the artworld changes over time. Not only do different agents have different roles within art institutions, but there is no need to presuppose consistency for such roles over time, meaning that the same work may be accepted and rejected as artwork by agents that have deontic powers for such verdicts. Collective recognition may still occur among dissidents or could simply change. Nevertheless, if the institutional structure is to have a say on what is an artwork, it must have a say on which declarative acts are successful or not in declaring works as artworks.

A final worry, commonly raised against institutional theories, is the problem of the isolated artists, *viz*., the possibility for a socially isolated artist to produce genuine artworks that are not within the scope of an institution. Some are worried that this is a too strong condition, as it seems counter-intuitive to deny the isolated artist the ability to produce art. But for advocates of the institutional account, it is not. Abell bit the bullet: the work of an individual operating independently from an institution is not an artwork. But as she claims, this is only problematic if by institution is meant a formalized and encoded set of rules and authority. On the contrary, following Searle's works, ''institutions do not require explicitly encoded rules, but merely regularities in the collective assignment of status function'' (Abell, 2012, p.687).

It should be added to Abell's response that the institutions involved in the artworld are diverse in their scope and structure. The São Paulo Museum of Art (MASP) and the Castelli Gallery in New York are formally encoded institutions that are included in the artworld. On the other hand, some of the status functions are performed by constitutive rules that are not formalized but still play a role in the institutional character of art. For example, enclosing a drawing with a frame or using colored paint are distinctive rules for making a painting. An isolated individual adopting such rules can be said to *ipso facto* be attempting to make a painting and can successfully be said to do so. In fact, we can say that (1) he will be performing an act that has the logical structure of a declaration, (2) he intends with his act to produce a painting,

[25] Going even further, there are artworks within the conceptualism tradition where there is no clear material medium to be presented, but that are legitimated as genuine artworks by the artistic community nonetheless. Such cases are famously called *dematerialized* art by Lippard (1973).

(3) the resulting product is accepted as a painting (by being enclosed by a frame and using colored paint, for example) and finally, (4) he has the powers to make a painting. In this case, the constitutive rules governing the making of a painting do impose restrictions on individuals who are apt in making one, it only happens that accepting the constitutive rules seems to be sufficient for granting one the power to do so.[26] With that in mind, the problem with the isolated artist is simply that, on a general level, there cannot be one. An individual completely isolated from every constitutive rule that is involved in the general concept of an artwork is almost impossible to find. And even if we could find such an individual, they would not be of much interest to us, being an alien to what we count as artistic.

In general, the institutional theory of art implies that what counts as being a painting is determined by the institutional structure as a whole, which explains why some institutions, such as museums, galleries, universities, and art critics, tend to have so much influence on the matter. The institutional theory of art also is precisely what the institutional account of authorship above captures when fixing the *F* description to *artworks*, where we can say that *w* is an artwork just in case *w* has an author according to conditions (1)–(4).

4 Can Artificial Agents be Authors?

Our goal is to answer the question of whether artificial agents can be authors in light of the institutional account of authorship just outlined. Also, our goal is to focus on cases coming from Generative Artificial Intelligence models in an artistic context. According to the conditions (1)–(4) of the institutional authorship account, to answer this question positively, such GAI systems require the following:

1. they must be able to perform acts that have the logical form of declarations and, accordingly, they must be able to express intentional states corresponding to beliefs and desires;
2. they must be able to meaningfully represent the content of its act as including the production of an artwork;
3. the resulting work must be accepted as an instance of an artwork, and finally
4. they must possess the required deontic powers to produce an instance of an artwork.

The first two conditions relate more directly to the intentional capabilities of these models, whereas the latter two are about the current institutional position of these models within the artworld. In other words, conditions (1) and (2) are about the possibility of GAI systems being considered genuine intentional agents, and conditions (3) and (4) relate to whether they can be accepted by the artistic community, viz., as institutional agents.

[26] For instance, if one wants to make a disruptive painting by rejecting most or all the constitutive rules governing the making of a painting, the artist must at least be an already recognized figure within the artworld, just as it was the case for Marcel Duchamp before the Fountain. One does not change the rules without the ability to be taken seriously. But one cannot be taken seriously without playing by the rules beforehand.

4.1 Intentional Agents?

Since at least the 1980s, the dominant perspective on AI has been the rational agency approach, where the goal is to develop systems that are capable of acting in a rational manner. Rao and Wooldridge (1999) define rational agents as being autonomous, reactive, and proactive in an environment (either physical or virtual). They are able to balance perception, deliberation, and action, and balance self-interest with community interest. Most AI systems using this approach have been built with specific and limited tasks in mind, and it has been the dominant approach for machine learning. A similar definition is found in Russell and Norvig's seminal textbook (2022, p.54): ''An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.''[27] What the agent perceives through its sensors is called a *percept*, and a *percept sequence* is the series of percepts accumulated over time. An agent's actions are then calculated using the agent's percept sequence and on built-in knowledge that it has, generating an *agent's function*, mapping percepts to actions. To implement an agent function is to build an *agent's program*. We can then say that an artificial agent is any agent implemented through an agent's program. Finally, an agent is intelligent if, for each percept sequence, the agent selects the course of action that ''maximize its performance measure'' (Russell and Norvig, 2022, p.58).[28]

Notice that in both the examples, there is no mention of a central feature of agenthood, viz., intentionality. The standard view of action and agency takes intentionality as a core component, according to which to act is to act intentionally:

> […] a being has the capacity to act intentionally just in case it has the right functional organization: just in case the instantiation of certain mental states and events (such as desires, beliefs, and intentions) would cause the right events (such as certain movements) in the right way. (Schlosser, 2019)

Still, there have been attempts in Artificial Intelligence since the 1980s to represent intentional states such as desires, beliefs, and intentions in artificial systems. One account of agency commonly used in the rational agency approach is the one advanced by Bratman (1987). In Bratman's theory of action, an agent weighs between his beliefs and desires to commit to a course of action. Thus, intentions are those desires that an agent *commits* to, that is, the desires he chooses to act upon. Based on this principle, a family of logics known as BDI-logics (BDI standing for belief, desire, and intention) has been proposed (See Meyer et al., 2015) having some successful implementations in the past.[29] In general, beliefs, desires, and intentions can be formalized using temporal parameters, as beliefs are the states of

[27] Other examples are Barandiaran et al. (2009) and Floridi (2021), who defines agents as transition systems showing interactivity, autonomy, and adaptability.

[28] The definition still seems too broad to be compared to human cases. And the case can be made that even this definition does not entail any relevant sense of intelligence, as argued by Floridi (2023). As he claims, the large language model GPT seems capable of having agency but still lacks any relevant conception of intelligence.

[29] See Wooldridge (2000) and Herzig et al. (2017) for a recent survey on BDI architectures and their challenges.

affairs that an agent believes to be true in the present time; desires are those states of affairs that an agent wants to bring about in the world in the future, and intentions are those desires that an agent has committed to by choosing a course of action.[30]

Recall that an agent is the author of an artwork *w* if (among other conditions) it can perform an act with the logical structure of a declarative. This structure includes that the agent needs to *believe* that *w* is an artwork, and *want w* to be an artwork (or any *F* that is institutionally dependent). Thus, the possibility of a program performing such action depends on the possibility of genuinely representing these intentional states with these specific contents. The common philosophical position considered in the rational agency AI comes from Dennett's intentional stance.[31] According to Dennett, the intentional stance or strategy consists of ''treating the object whose behavior you want to predict as a rational agent with beliefs and desires and other mental states'' (Dennett, 1981, p.59). Objects whose behavior can be predicted this way are called *intentional systems*. The intentional stance is useful in predicting behavior when there's no clear way to predict how a system will behave in terms of its physical features or the way it was designed.[32] Dennett's point is that there is no need to equate the attribution of an intentional state to a mental entity, somewhat located in the agent's brain. For Wooldridge (2000), the intentional stance is nothing more than an abstraction tool, a ''convenient shorthand for talking about complex systems'' that is useful for computer scientists to formalize and implement these behaviors. This goes in line with John McCarthy's (1979) famous position that ascribing mental states to a machine is useful to understand the machine's structure, whereas even saying that thermostats have beliefs about the room temperature is legitimate. In any of the options above, a GAI system would easily satisfy condition (1) of the definition, provided the correctness of the intentional stance for the relevant cases.

Nonetheless, the extent to which the formalization and implementation of intentional states (such as beliefs, desires, and intentions) correlates to the same intentional states in humans is not a settled matter. Consequently, it is questionable whether these implementations satisfy condition (1). Declarations, as speech acts, are bounded by the social and institutional contexts in which they are performed, a *world of utterance* in Searle and Vanderveken's terms (1985). Assuming authorship to be primarily a relation defined over the same social and institutional contexts, this strategy seems insufficient. Even granting that the intentional stance is useful for most complex system, it still does not apply equally in all cases. For instance, we don't know how human behavior is conditioned by its internal states, and similarly,

[30] They can be then paired with possible-world semantics, having branching temporal structures as worlds. See, for example, Wooldridge (2000).

[31] We can find the use of Dennett's philosophy in Wooldridge (2000).

[32] Dennett (1981, p.61) describes the strategy in the following steps: ''first you decide to treat the object whose behavior is to be predicted as a rational agent; then you figure out what beliefs that agent ought to have, given its place in the world and its purpose. Then you figure out what desires it ought to have, on the same considerations, and finally you predict that this rational agent will act to further its goals in the light of its beliefs. A little practical reasoning from the chosen set of beliefs and desires will in many - but not all - instances yield a decision about what the agent ought to do; that is what you predict the agent will do.''.

we don't know how the stock market's behavior is determined by all the relevant institutions and agents involved. Although the possibility to take an intentional stance on both cases remains, saying that ''John believes it will rain'' and ''The market believes there will be a recession'' involves two different ascriptions of beliefs, as the first is taken to be literal, while the latter is a metaphorical instance of the former. Likewise, in saying that Amazon's Alexa believes it will rain, the intentional stance is useful to predict Alexa's behavior just as it is to predict John's. But it does not entail that John and Alexa believe in the same way, as Alexa does not have the commitments that John has for the same belief, e.g., carrying an umbrella to work.[33] Using the intentional stance may be useful both in human and artificial agent cases, but the differences in how we ascribe and understand intentionality on both may persist.

For all intents and purposes, GAI systems satisfy the above definitions of artificial agents: they are reactive systems designed to interact with a controlled environment, can learn from a dataset through Machine Learning techniques, and can perceive its environment (through prompt commands), interpret these percepts and react accordingly. But can they also be intentional agents? They are highly complex agents designed as multilayered artificial neural networks. A multilayered neural network is composed of input layers, an output layer, and a stack of hidden layers, each having multiple neurons. Each layer transforms the data received in the input layers, passing it along subsequent layers of neurons until it reaches the output layer.[34] Although the training is a finite and computable process, there is no feasible way to completely know all the transformations in the hidden layers, as complex GAI models can have up to hundreds of layers. GPT3.5 is assumed to have 96 layers and up to 175 billion parameters.[35] It is then useful to adopt the intentional stance since a clear understanding of how their neurons act is not available. However, claiming that GPT or Midjourney has beliefs or desires is still not the same as saying that a human agent has beliefs and desires.

Returning to the authorship problem, Midjourney is in fact causally responsible for *Theatre d'Opera Spatial* having the visual characteristics that it has. But is it intentionally responsible? Even assuming the intentional stance, the answer would depend on whether the ascription of beliefs and desires to Midjourney is *mutatis mutandis* the same as in human cases. A stronger and negative answer was provided by Searle (1980) with the famous Chinese Room experiment. His point is that intentionality results from causal features of the brain that cannot be simulated in a program, meaning that ''Any attempt literally to create intentionality artificially (strong

[33] The differences are even greater concerning ethical cases. van Woudenberg et al. (2024) argue that a moderate view on authorship, based on Dennett's intentional stance, cannot account for the normative significance of our practices, meaning that e.g., ChatGPT cannot take moral responsibility for their actions without having real intentional states (for a contrary view, see Floridi (2021)). A similar point is made by Arora (2024), that machines cannot be responsible for their assertions given their capacity to only make proxy assertions.

[34] See Foster (2023), Chap. 2, on the basics of deep learning techniques.

[35] A parameter is roughly a variable that is adjusted during the training process. As of the time of this writing, OpenAI has not yet disclosed GPT4's size in terms of layers and parameters, although speculation estimates more than 100 layers and up to a trillion parameters.

AI) could not succeed just by designing programs but would have to duplicate the causal powers of the human brain.'' (Searle, 1980, p.417) Searle's argument could be answered by taking modern GAI systems, based on deep neural networks, as literal attempts to duplicate the human brain, not the human mind. But even then, it is hard to avoid the point that LLMs (such as GPT) are nothing more than stochastic parrots[36] and cannot properly understand text prompts or their corresponding outputs since they lack semantics.[37] We can assert, following Searle, that artificial agents lack genuine semantic understanding of their data, as they cannot go beyond their syntax. As he puts it, ''All the computer has is more symbols''. (p.423) Likewise, GAI systems such as Midjourney and Dall-E are nothing more than stochastic drawing bots trained to manipulate images as sets of pixels, not artworks.

We may recall conditions (1) and (2) as questions:

(Q1) Is *Theatre d'Opera Spatial* the result of an act performed by Midjourney with the logical form of a declarative?
(Q2) Does it have the content tha *Theatre d'Opera Spatial* is an artwork?

If we took a Searlean route, then the possibility of artificial agents having intentional states is denied, and so is condition (1) of the institutional authorship definition, meaning that (Q1) is answered negatively. In other words, artificial agents cannot fulfill the conditions for expressing illocutionary force and, therefore, cannot be speech actants. On this issue, Nickel (2013, p.499) argues that some systems of natural language generation can be taken to be speech actants and to make authorial contributions to a message, given that they can express grammatical generativity, draw on some background human rationality and show sensitivity to situational features of language use. But since they lack a normal understanding of agency and intentionality, their speech only has *prima facie* force, or apparent force, since it is not clear whose force it is. If so, then artificial agents are only capable of proxy speech. Is proxy speech enough? Nickel relies on Dennett's intentional stance: not having intentions ''is not a barrier to the hearer's reconstruction of speaker meaning'' (p.499). However, ethical responsibility is still out of bounds for artificial agents. As Nickel concludes, proxy speakers have limited autonomy and, thus, limited responsibility. In a recent paper, Arora (2024) offered a similar but more precise argument for proxy assertions in machines. In both cases, the core presupposition is that machines lack intentional states and, therefore, cannot be speech actants. The same poins is made *mutatis mutandis* by Gubelmann (2024). In all cases, the underlying conclusion seems the same: with no proper intentionality, there can be no genuine speech act.

[36] The term was first used in Bender et al. (2021). It implies that large language models such as GPT lack real language understanding, as they are ''not grounded in communicative intent, any model of the world, or any model of the reader's state of mind.'' (p.616).

[37] In Searle's words, ''formal symbol manipulations by themselves don't have any intentionality; they are quite meaningless; they aren't even symbol manipulations, since the symbols don't symbolize anything. In the linguistic jargon, they have only a syntax but no semantics. (Searle, 1980, p.422) This is also known as the *symbol grounding problem*, following Harnad (1990).

Could the same be said about authorship? According to the intentional stance, (Q1) could be answered positively. Either the system represents explicitly in its syntax something that resembles intentional states (such as the implementations of BDI-logics) or, in the case of deep learning architectures, it somewhat represents internally in its layers and billions of parameters something similar to beliefs and desires. In such case, Midjourney somewhat *believed* and *desired Theatre d'Opera Spatial* to be an artwork, and so it is possible for a GAI system to perform an act that somewhat has the logical form of a declarative. The resulting argument would have the following structure with a weakening condition:

(P1) If it looks like an agent has intentional states such as beliefs and desires (*per* intentional stance), then it looks like it can perform declarative acts.
(P2) If it looks like it can perform declarative acts, then it looks like it can be an author (*per* institutional authorship definition).
(C) Therefore, if it looks like it can have intentional states, then it looks like it can be an author.

But this argument depends on the condition of authorship being weak author-relevant in *all* cases, following Uidhir's definition of author-relevance. Cases where authorship is strong author-relevant demands that an *F*-work cannot just look like to have an author, it effectively has to have one. In the case of institutional authorship here explored, this means that an agent has to effectively be capable of performing declarative acts and *a fortiori* being capable of having intentional states. In effect, the weak author-relevance condition must be added to (P2):

(P2') If it looks like it can perform declarative acts, then it looks like it can be an author (*per* institutional authorship definition) of a weak author-relevant *F*-work.

Which will render the new conclusion (C'):

(C') If it looks like it can have intentional states, then it looks like it can be an author of a weak author-relevant *F*-work.

Thus, whether the intentional stance is sufficient for rendering an artificial agent the author of a *w* as an *F* is dependent also on the *F*-type, meaning that (Q1)'s answer depends on (Q2), that is, on the agent's capacity of effectively representing the *F*-type, *per* the necessary requirements for *F*'s. In sum, even accepting the intentional stance, authorship would be dependent on the *F*-type being considered to be authored, meaning that we must look into the content of the act, and thus, to (Q2).

Ideally, (Q2) could be answered positively. An act performed by an artificial agent can, in principle, be programmed to have as content that its output is an artwork. However, the problem concerns some types of *F* descriptions, particularly the ones that we are interested in here: the *F*'s that are institutionally relevant, such as being an artwork. For some descriptions *F*, an agent cannot genuinely intend to do a *F* without having some relevant experience of *F*'s. For example, an agent cannot intentionally write a verdict without, to some extent, having experience of verdicts,

viz., institutional understanding of the legal system in which verdicts are defined. But, having a description of the legal system does not suffice for institutional experience. For such, an agent must be an effective participant in the institutional reality, and thus, all *F*'s that are institutionally dependent are strong author-relevant in Uidhir's (2013) sense. It is not sufficient to have only *prima facie* institutional experience to be considered a participant of the institutional reality.

Connecting conditions (1) and (2) for institutional *F*'s requires that an agent can only meaningfully intend to produce an *F* if he/she has relevant experiences (viz., institutional) of *F*'s. Following the institutional theory of art, having real and institutional experience of artworks is a necessary condition for representing artworks in a declarative act and intending to produce artworks. This is hardly achievable for artificial agents, even assuming the intentional stance. Granting that Amazon's Alexa has the belief that it is raining does not entail that Alexa has any relevant experience of rain.[38] Similarly, even if we can represent intentional states in a formal model adopting the intentional stance, they would lack the institutional competence required for understanding the sortal description *artwork* in a meaningful way, as they don't have genuine experience of artworks as institutional entities. Granting that Midjourney looks like it has the belief and desire to produce *Theatre d'Opera Spatial* as an artwork does not entail that it has any real experience of artworks. And since it only looks like it is producing one, it cannot be a genuine author. In conclusion, even in the intentional stance scenario, the possibility of answering positively (Q1) and (Q2) is dependent on Midjourney's ability to be an institutional agent.

4.2 Institutional Agents?

According to conditions (3) and (4) of the institutional authorship definition, in order to an artificial agent *A* be the author of *w* as an *artwork*, *w* must be accepted as an artwork by the relevant institutions, i.e., the artworld, and agent *A* must possess the required deontic powers to realize an instance of an artwork, relative to the artworld. According to the institutional theory of art, what counts as artwork is conditioned by the institutional framework to which the type description *artwork* is relevant. Thus, per condition (3), there is *ideally* no difference in evaluating the acceptance of *w* that is realized by an artificial agent to the same *w* being realized by a human agent. Either *w* is accepted, or it isn't, as an artwork solely by the features that *w* has[39]. Returning to the Midjourney case with *Theatre d'Opera Spatial*, the question

[38] Following Russell's (1905) distinction, we can say that Alexa knows descriptions but cannot have a proper acquaintance of rain events.

[39] The problem is that *w* has both intrinsic and relational features. It is possible, and even common, to accept a given object *w* as an *artwork* precisely because *w* is already known to be authored by a recognized artist (that is, accepting *w* as an *artwork* due to its relational features). This is not a necessary condition (although it can be a sufficient one). It seems that *Theatre d'Opera Spatial* was accepted by its intrinsic features, but we can still imagine that the jury expected it to be authored by a human. There is no way to know a priori which features (either intrinsic or relational) are going to be relevant. Although we say that *ideally* it is possible to evaluate an artwork in terms of its intrinsic features, other cases are not excluded. The important point is that for condition (3), only features of the object are considered (intrinsic or not), whereas for condition (4), features of the agent/author are the relevant ones. (I am grateful to an anonymous referee for pointing out this issue.)

(Q3) Is *Theatre d'Opera Spatial* accepted as an instance of an artwork?

can be answered positively. Practical examples were already mentioned. *Theatre d'Opera Spatial* did won the 2022 Colorado State Fair's annual art competition. In the 2023 Prêmio Jabuti case, officials disqualified an artwork made by Midjourney only after realizing that the name ''midjourney'', which appeared in the work's entry for the award, was not a human. Nothing in the visual features of both works made them unworthy of being ruled as artworks.

To understand how conditions (3) and (4) relate, consider the case where *w* is a disruptive case. We consider a *w* to be disruptive if *w* is an attempt to be an *F* but includes features (or lack of) that are not accepted for an *F* relative to the institutions governing *F*'s. Disruptive attempts are usually met with resistance by the *F*-community, and there are plenty of examples in the history of art. Duchamp's *Fountain*, at the time of its exhibition, should have failed condition (3). John Cage's famous silent music piece *4'33''* in the 1950s would also fail condition (3) on the grounds that silence cannot be considered a musical work. But both cases were embraced within the artworld since both Duchamp and Cage were in a position to be taken seriously (Duchamp was already a recognized artist, and Cage was a trained composer). Successful disruptive cases always emerge from within the relevant institutions because condition (4) must be somehow satisfied in order for the *F* description to be changed.

Conversely, a work *w* can be accepted as an artwork by satisfying condition (3) even without satisfying condition (4) beforehand. Think about a debuting artist. In order to be accepted by the artistic community, there must be some artwork being presented. One cannot be an artist without artworks. Hence, to be a debuting artist, condition (3) is mandatory.[40] One cannot be accepted as an artist by failing to meet both conditions (3) and (4) (a debuting artist could hardly have been the author of *Fountain*). Considering that institutions, the deontic powers, and the conditions for satisfying the *F*-type are contingent, the debuting artist can eventually be accepted (satisfying (4) by satisfying (3) beforehand), and the disruptive work can be recognized as an artwork (satisfying (3) by satisfying (4) beforehand).

The current problem with artificial agents is that condition (4) seems out of their bounds. *Theatre d'Opera Spatial* can satisfy the conditions for being an artwork, but without condition (4), Midjourney cannot be its author. The question deriving from condition (4) is:

(Q4) Does Midjourney possess the required deontic powers to produce an instance of an artwork?

[40] One can argue against condition (3) by claiming that a debuting artist is usually accepted in the art community by producing innovative artworks that do not conform to standard features expected for artworks. However, there is a difference in considering *w* as a *new kind of artwork* to *not an artwork*. Disruptive cases usually start as the latter, and only after some acceptance (by satisfying condition (4)) can they be considered as the former.

There are two ways to answer this question: The first one is to examine whether Midjourney and other GAI systems *currently* possess such deontic powers. The second one is to answer whether they *could* possess them. On the first, it is currently not the case that such systems possess deontic powers within the art institutions. If they did, there would be no concerns with copyright claims for *Theatre d'Opera Spatial*, and the Midjourney's Frankstein illustration would not be disqualified from Prêmio Jabuti. The controversy surrounding GAI systems and their products is a testament to their current lack of deontic powers within the artistic community.

But the more important point is whether they *could* possess them. Examples of deontic powers involved in the institutional reality are ''duties'', ''permissions'', ''rights'', ''obligations'', etc. According to Searle (2010, ch.6) they are ''desire-independent reason for action''. By desire-independent, Searle means those reasons for action that an agent does not necessarily have the desire to fulfill. These desire-independent reasons are constituted into a system of deontic powers. They are connected with some modal auxiliary verbs such as ''ought'', ''should'', ''can'', ''must'', etc., that indicate the desire-independent reasons for action and form the basis of our institutional reality. Searle points out that one crucial condition for deontic powers to fulfill their roles is the *causal gap*. It is worth quoting him in full:

> we have a especial kind of consciousness in which we have a sense of making decisions that are not forced, the kind of consciousness where we choose one thing but we have a sense that we could have chosen something else. In such cases we sense a *causal gap* between the reasons for our decisions and actions and the actual decisions and actions. […] without the gap — that is, without the consciousness of freedom — institutional structures are meaningless; but with the gap, they are essential. (Searle, 2010, p.133)

If my actions were completely determined by my reasons for acting, there would be no causal gap between them. There would be no deontology since ''duties'', ''permissions'', ''rights'', ''obligations'' would be meaningless, as there is no sense in which I ought, should, can, or must do something if, in principle, I will do it anyways. Thus, the presupposition of a causal gap between reasons and actions is necessary for the structure of institutional reality to be of any use and to have the structure that it has.

If Searle's argument is correct, the possibility of artificial agents possessing deontic powers is damned, as there is no gap between reasons and actions performed by them. The fact that we don't know the details of why a GAI system generates a particular image, text, or video (as we cannot track these decisions on the vast network of layers and neurons in their architecture), does not presuppose the existence of a gap. A GAI system does not *opt* in generating a particular image rather than another. It simply generates content according to the dataset and some randomness. Although human behavior is reason-driven, one of its key features is the possibility of bypassing them. I can have reasons and desires to act but ultimately choose not to. This feature does not seem to be present in the behavior of artificial agents. They only act as they are supposed to act.

Can artificial agents like robots, virtual assistants, and GAIs make promises, buy properties, order a pizza, or declare a session to be opened? They can be programmed to behave in a similar way. A robot can say ''I promise'', make a deposit, ask for a pizza, or start a session. But these are only mechanical behaviors attempting to imitate humans. There is no real sense of rights, duties, or obligations involved simply because there is no causal gap. ''They are programmed to respond in certain ways, and that's that.'' (Searle, 2010, p.136). Since there is no causal gap, there are no deontic powers. Since there are no deontic powers, there is no reasonable sense in which they can be part of the institutional reality.

In conclusion, condition (4) cannot be satisfied by an artificial agent. There is no reasonable sense in which, e.g., Midjourney can be said to have the right to produce an instance of an artwork. It can, in fact, produce something that resembles an artwork, as already stated. But it does not imply that Midjourney has authored *Theatre d'Opera Spatial*, as authorship requires relevant participation in institutions that govern our understanding of the description *artworks*. Thus, (Q4) can only be answered negatively. And since Midjourney or artificial agents, in general, cannot be institutional agents, they cannot have the required experience of *F*-types that are institutionally dependent, meaning that they cannot truly intend to produce artworks, thus also answering (Q1) and (Q2) negatively.

5 Conclusion

Artificial agents such as the generative artificial intelligence systems cannot be authors of artworks or any *F*-type description that is institutionally dependent, given that they fail conditions (1), (2), and (4) of the institutional authorship definition. The reason behind this conclusion, as stated above, is the following: since artificial agents cannot be institutional agents, they fail condition (4). Failing condition (4) means that they cannot properly understand the *F*-type descriptions that are institutionally relevant, meaning that they fail condition (2). Given that they fail condition (2), they cannot properly intend to produce artworks or any *F*-type description that is institutionally dependent, regardless of the intentional theory considered. In other words, consider van Woudenberg et al. (2024) distinction between conservative, moderate, and liberal views on authorship. In a conservative view, artificial agents cannot have intentional states, and thus, they fail condition (1). In a moderate view in which the Intentional Stance is considered, they can have similar states to human intentions. However, they would not matter for authorship, given that by failing conditions (4) and (2), an agent cannot intend to produce something they cannot experience in a meaningful way. Finally, a liberal view accepts that artificial agents can have intentional states and can be authors. However, accepting the liberal view can make authorship a trivial relation, as human and non-human productions would equally be candidates to be authored.

Condition (3), however, is still possible, as it relies solely on the features of the *w* object satisfying the *F* description. This leaves open the possibility for human appropriation of the GAI systems outputs into their own works. A corollary of the institutional authorship definition is the lack of necessary causal connections between an author and their artwork. This is particularly the case for ready-mades

and appropriation cases. To *A* be an author of *w* as an artwork, it is sufficient for *A* to (1) act in conformity with a declaration, that (2) has as content that *w* is an artwork, with (3) *w* being accepted as an artwork and (4) *A* possessing the deontic powers for declaring *w* an artwork.

Some may consider this a problem for the institutional definition, as an agent could simply declare an already existing artwork to be his/her by an act of declaration. However, these cases are readily blocked by the institutional understanding of the cases open to be declared, meaning that objects that already have a declared author cannot be reclaimed. There are, however, cases of appropriation art that are still recognized as artworks.[41] This happens because the declarative act in question is not properly an appropriation of an already existing artwork but the realization of a new artwork that has as content the appropriation of another.[42] Under this explanation, appropriation cases are not properly speaking cases of genuine appropriation.[43] It is also important to notice that the very act of declaration is an already authorship-biding act, viz., to declare is to *ipso facto* claim authorship. I cannot declare a war, a marriage, or a session to start without having as a consequence that I was responsible for the war, the marriage, or the session to have started. The speaker of a declaration is the author of the declaration *simpliciter*.

Lastly, recall that artist Jason Allen did try to claim *Theatre d'Opera Spatial* to be his artwork. To be successful, he needed to (1) make a declaration that (2) has as content the claim that *Theatre d'Opera Spatial* is an artwork, (3) *Theatre d'Opera Spatial* had to be accepted as artwork, and (4) he needed to be allowed to perform the act he was performing. Conditions (1), (2), and (3) seem true at current artworld standards. But condition (4) is still debatable. It was based on not meeting condition (4) that the Copyrights Office rejected his claim of copyrights, not granting him the necessary deontology. But although resistance is expected at the present time, the contingent character of the institutions means that restrictions can be overcome in the future.

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[41] See Uidhir (2013, ch.2) for some examples and discussion.

[42] We can call these cases second-level declaratives, since their propositional content involves other uttered declarations.

[43] Uidhir (2013, p.59) similarly argues that the intentions of an original artwork do not figure on its appropriation work, meaning that ''authorship cannot be inherited merely via appropriation''.

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